Data Cleaning

```
# Step 7: read csv file
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ---------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(skimr)
library(lubridate) # for date changing

file_path <- "data/Airbnb_Open_Data_cleaned_first.csv"

df_cleaned_first <- read_csv(file_path) %>%
  clean_names()
```

```
## Rows: 102058 Columns: 23
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (9): NAME, host_identity_verified, host name, neighbourhood group, neig...
## dbl (13): id, host id, lat, long, Construction year, price, service fee, min...
## lgl  (1): instant_bookable
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(df_cleaned_first)
```

```
## # A tibble: 6 x 23
##        id name      host_id host_identity_verified host_name neighbourhood_group
##     <dbl> <chr>       <dbl> <chr>                  <chr>     <chr>
## 1 1001254 Clean & ~ 8.00e10 unconfirmed            xiayi     Brooklyn
## 2 1002102 Skylit M~ 5.23e10 verified               Jenna     Manhattan
## 3 1002403 THE VILL~ 7.88e10 <NA>                   Elise     Manhattan
## 4 1002755 blank     8.51e10 unconfirmed            Garry     Brooklyn
## 5 1003689 Entire A~ 9.20e10 verified               Lyndon    Manhattan
## 6 1004098 Large Co~ 4.55e10 verified               Michelle  Manhattan
## # i 17 more variables: neighbourhood <chr>, lat <dbl>, long <dbl>,
## #   instant_bookable <lgl>, cancellation_policy <chr>, room_type <chr>,
## #   construction_year <dbl>, price <dbl>, service_fee <dbl>,
## #   minimum_nights <dbl>, number_of_reviews <dbl>, last_review <chr>,
## #   reviews_per_month <dbl>, review_rate_number <dbl>,
## #   calculated_host_listings_count <dbl>, availability_365 <dbl>,
## #   house_rules <chr>
```

```r
# Step 8 & 9: type transform and missing value

# median
median_reviews_per_month <- df_cleaned_first %>%
  pull(reviews_per_month) %>%
  median(na.rm = TRUE)

median_availability <- df_cleaned_first %>%
  pull(availability_365) %>%
  median(na.rm = TRUE)

# operation
df_cleaned_second <- df_cleaned_first %>%
  mutate(
    # mutate value type
    price = as.numeric(price),
    service_fee = as.numeric(service_fee),

    reviews_per_month = replace_na(reviews_per_month, median_reviews_per_month),

    availability_365 = replace_na(availability_365, median_availability),

  )

# check for missing value
skimr::skim(df_cleaned_second)
```

```
## Warning: There was 1 warning in `dplyr::summarize()`.
## i In argument: `dplyr::across(tidyselect::any_of(variable_names),
##   mangled_skimmers$funs)`.
## i In group 0: .
## Caused by warning:
## ! There were 3703 warnings in `dplyr::summarize()`.
## The first warning was:
## i In argument: `dplyr::across(tidyselect::any_of(variable_names),
##   mangled_skimmers$funs)`.
## Caused by warning in `grepl()`:
```

```
## ! unable to translate 'Williamsburg?<80>?Steps To Subway, Private Bath&Balcony' to a wide string
## i Run `dplyr::last_dplyr_warnings()` to see the 3702 remaining warnings.
```

Table 1: Data summary

Name	df_cleaned_second
Number of rows	102058
Number of columns	23
Column type frequency:	
character	9
logical	1
numeric	13
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
name	0	1.00	1	251	0	61250	0
host_identity_verified	289	1.00	8	11	0	2	0
host_name	0	1.00	1	35	0	13148	0
neighbourhood_group	29	1.00	5	13	0	5	0
neighbourhood	16	1.00	4	26	0	224	0
cancellation_policy	76	1.00	6	8	0	3	0
room_type	0	1.00	10	15	0	4	0
last_review	15832	0.84	8	10	0	2477	0
house_rules	51842	0.49	6	1004	0	1964	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
instant_bookable	105	1	0.5	FAL: 51186, TRU: 50767

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
id	0	1	2.918438e+07	1.627173e+07	1001254.00	1.509286e+07	2.918438e+07	4.327590e+07	5.736742e+07	
host_id	0	1	4.926738e+10	2.863742e+10	2600518.00	2.459924e+10	4.912865e+10	7.400619e+10	9.876313e+10	
lat	8	1	4.073000e+01	6.000000e-02	40.50	4.069000e+01	4.072000e+01	4.076000e+01	4.092000e+01	
long	8	1	-7.395000e+01	5.000000e-02	−74.25	-7.398000e+01	-7.395000e+01	-7.393000e+01	-7.371000e+01	
construction_year	214	1	2.012490e+03	5.770000e+00	2003.00	2.007000e+03	2.012000e+03	2.017000e+03	2.022000e+03	
price	247	1	6.253600e+02	3.216700e+02	200	3.400000e+02	6.220000e+02	9.120000e+02	1.200000e+03	
service_fee	273	1	1.250400e+02	6.423000e+01	100	6.800000e+01	1.250000e+02	1.820000e+02	2.400000e+02	
minimum_nights	400	1	8.150000e+00	3.034000e+01	1.00	2.000000e+00	3.000000e+00	5.000000e+00	5.645000e+03	
number_of_reviews	183	1	2.752000e+01	4.957000e+01	0.00	1.000000e+00	7.000000e+00	3.000000e+01	1.024000e+03	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
reviews_per_month	0	1	1.280000e+00	1.620000e+00	0.01	2.800000e-01	7.400000e-01	1.710000e+00	9.000000e+01	
review_rate_number	319	1	3.280000e+00	1.290000e+00	1.00	2.000000e+00	3.000000e+00	4.000000e+00	5.000000e+00	
calculated_host_listings_count	319	1	7.940000e+00	3.027000e+01	1.00	1.000000e+00	1.000000e+00	2.000000e+00	3.320000e+02	
availability_365	0	1	1.408500e+02	1.351600e+02	0.00	3.000000e+00	9.600000e+01	2.618000e+02	3.677000e+02	

```
# Step 10: change date

df_cleaned_second  <- df_cleaned_second  %>%
  mutate(
    last_review = lubridate::mdy(last_review) # use mdy()
  )

# define date limit
current_date_limit <- as.Date("2021-01-01")

# calculate median date
median_review_date <- df_cleaned_second  %>%
  filter(!is.na(last_review) & last_review <= current_date_limit) %>%
  pull(last_review) %>%
  median(na.rm = TRUE)

# change
df_cleaned_second  <- df_cleaned_second  %>%
  mutate(
    # if last_review is NA or date exceeds limit, use median instead
    last_review = if_else(is.na(last_review) | last_review > current_date_limit,
                          median_review_date,
                          last_review)
  )

# check date
summary(df_cleaned_second$last_review)
```

```
##         Min.      1st Qu.       Median         Mean      3rd Qu.         Max.
## "2012-07-11" "2019-01-02" "2019-05-20" "2019-01-02" "2019-06-15" "2021-01-01"
```

```
# Step 11: duplicate data handling

df_cleaned_second <- df_cleaned_second %>%
  # set minimum_nights
  mutate(
    minimum_nights = case_when(
      # no <0
      minimum_nights <= 0 ~ 1,
      # no outlier
      minimum_nights > 365 ~ 365,
      # everything else keep the same
      TRUE ~ minimum_nights
    )
  )
```

```r
# check lines
print(paste("lines after duplicate data handling", nrow(df_cleaned_second)))
```

```
## [1] "lines after duplicate data handling 102058"
```

```r
# check
summary(df_cleaned_second$minimum_nights)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   1.000   2.000   3.000   7.968   5.000 365.000     400
```